Q18. Directory structure and file management

Original question: Create and manage a directory structure in Kali Linux. Perform file creation, copying, renaming, displaying, and deletion operations. Verify the final directory contents and document the procedure.

Aim

Create a small directory structure, write/read a file, copy and rename it, delete the designated original and verify final contents safely.

Requirements

Kali/Linux terminal and an ordinary writable home directory. This practical uses nested source/archive folders, unlike Q1's single directory.

Procedure

Step 1 - Create a fresh root and child folders

```
cd "$HOME"
mkdir EHF_Structure
cd EHF_Structure
mkdir source archive
pwd
ls -l
```

If EHF_Structure already exists, inspect it and choose a fresh lab workspace with the examiner. Do not silently overwrite earlier files. `mkdir source archive` creates two sibling subdirectories.

Step 2 - Write, display and copy

```
printf '%s\n' 'Practice evidence - Q18.' > source/report.txt
cat source/report.txt
cp -i source/report.txt archive/report_copy.txt
ls -l source archive
```

The paths are relative to EHF_Structure. `>` writes a new file (and can truncate an existing one); `cat` verifies the text; `cp -i` asks before overwriting a destination. Parent directories must already exist.

Step 3 - Rename the archived copy and compare

```
mv -i archive/report_copy.txt archive/report_final.txt
cmp source/report.txt archive/report_final.txt
printf 'Comparison exit status: %s\n' "$?"
cat archive/report_final.txt
```

The rename is inside archive. A comparison status of 0 confirms identical content before the original is deleted; it does not prove filesystem metadata are identical.

Step 4 - Delete only the practice original and verify

```
pwd
ls -l source/report.txt
rm -i source/report.txt
ls -la source
ls -la archive
find . -maxdepth 2 -type f -print
```

Confirm the path, then answer y to the rm prompt. `find` lists regular files up to two levels below the current directory. Expected final state: source remains empty and archive contains report_final.txt. Avoid wildcards, rm -rf and sudo. Optional cleanup after marking: `rm -i archive/report_final.txt`, then `rmdir source archive`, `cd ..`, `rmdir EHF_Structure`.

Expected output - illustrative

```
Practice evidence - Q18.
Comparison exit status: 0
./archive/report_final.txt
```

This summarizes the text display, successful comparison and final find output. The ls listings also contain directory entries and metadata that vary by machine.

Observation and result

Record the absolute root, final directory structure and content of report_final.txt. Show that source/report.txt and archive/report_copy.txt are absent while archive/report_final.txt remains readable. The requested creation, copying, renaming, display, deletion and verification have been demonstrated in a nested structure.

Common errors

- cp cannot create destination: create archive first or verify your current path.
- Wrong file deleted: stop, retain the record and use any preserved copy; do not claim rm has an undo function.
- Overwrite prompt: inspect the existing destination instead of automatically answering y.

Viva questions

- Why use separate source and archive folders? They demonstrate relative paths and keep the retained copy clearly identified.
- What does -i do for cp, mv and rm? It enables interactive confirmation in the relevant overwrite/removal situations.
- Does rm move a file to the desktop trash? Normally no; shell rm unlinks it directly.
- Why verify before deletion? To ensure the intended copy exists and contains the expected bytes.